

Defocused image of a periodic complex object in an optical system under partially coherent illumination

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(Received 10 October 1975; revision received 10 May 1976)

Investigations have been carried out on the combined effects of amplitude and phase variations in a periodic complex object on the image formed in a partially coherent optical system in the presence of defocusing. To examine these effects, images of one-dimensional periodic complex objects and the two-dimensional Siemens Star are simulated with various different complex transmittances. A number of significant phenomena are clarified through observation of the simulated images and evaluation of the image contrast for the fundamental and second harmonic spatial frequency components. One of the results is that object phase variations are extremely important to the image appearance in the defocused system and to the change of contrast reversal frequency with defocusing under incoherent and partially coherent illumination, but not in the coherent case.

In two previous papers,^{1,2} we reported combined effects of amplitude transmittance and phase variation in a complex object on the irradiance in an image and its contrast in a diffraction-limited aberration-free optical system under conditions of partial coherence. To treat the stated problem, based on Fourier series analysis, we showed the general expression to obtain the image and image contrast for a periodic complex object in any illumination mode, and then we wrote a computer program to calculate them. Although numerical results associated with the aberration-free optical system gave significant information about partially coherent imaging of a complex object, it is not sufficient to describe the image characteristics fully. The reason is that, when we observe the image of a small specimen with thickness in an ordinary microscope, defocused images necessarily superimposed on the infocused image due to shallow focal depth of an objective. Therefore, it is important to know the characteristics of the defocused image of the complex object, because most semitransparent objects embedded in the transparent medium include the phase variations as well as variations in amplitude transmittance.

Although a number of papers³⁻⁸ have been presented on the defocused image in a partially coherent optical system, little work has been done for the complex object.

In this paper, by use of the algorithm which we made, we simulate the defocused image and calculate the image contrast for a periodic complex object in an optical system under six different illumination modes, and examine the effect of defocusing on the image characteristics for the complex object. The algorithm which we have used is applicable to obtain the image in a diffrac-

tion-limited optical system in the presence of aberration and defocusing by adjusting the pupil function adequately. First, we calculate the contrast of an image of a square gratinglike complex object in the defocused optical system under different conditions of partial coherence. Then, we obtain simulated images, and discuss combined effects of amplitude transmittance and the phase variation in a complex object on image appearances for the selected objects, through observations of simulated images. Finally, images of the Siemens star that has amplitude transmittance and phase variation, are simulated, and their characteristics are compared with the image contrast of the fundamental and second harmonic spatial frequency components obtained for the corresponding one-dimensional complex object.

I. EVALUATION OF IMAGE CONTRAST

Applying the computer program which we made previously to the present problem, we calculated the image contrast of the defocused image of the square gratinglike complex object that has amplitude and phase distributions, shown in Fig. 1(a). Amplitude transmittance is specified by parameters A and C , and the phase variation by the parameter B . In the present case, we set $A = 0.25$ and $C = 1$. The pupil function $f(x)$ can be described by

$$f(x) = \exp[i(2\pi/\lambda) D_F x^2], \quad (1)$$

where x is the reduced coordinate⁹ in the pupil plane, D_F is the wave aberration to specify the amount of defocusing, and λ is the wavelength of the illuminating light.

Figure 2 shows the image contrast [defined by Eq. (15) in

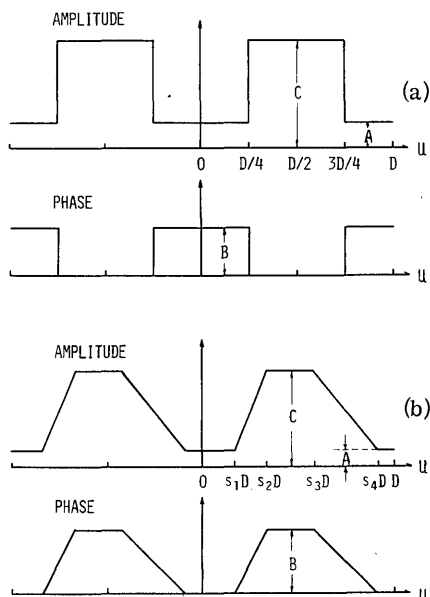


FIG. 1. Models of complex objects. (a) Square gratinglike complex object, and (b) trapezoidal gratinglike one.

Ref. 2] for the fundamental and second harmonic spatial frequency components in a defocused optical system under six different illumination modes, in which the sign of the image contrast for the fundamental spatial frequency component is reversed. In Fig. 2, the abscissa means the spatial frequency specified by the reduced coordinate and the ordinate is the image contrast defined by the ratio of the Fourier coefficient of k th harmonic spatial frequency component in the image irradiance to the bias. Conditions of partial coherence corresponding to individual figures are specified by the radiance distribution $\gamma(x)$ of the effective source. They are (a) $\gamma(x) = \delta(x)$ (central coherent illumination); (b) $\gamma(x) = 1$ for $|x| \leq 0.5$, and 0 otherwise; (c) $\gamma(x) = 1$ for $|x| \leq 1.0$, and 0 otherwise; (d) $\gamma(x) = 1$ for $0.95 \leq |x| \leq 1.0$, and 0 otherwise (annular illumination); (e) $\gamma(x) = x^2$ for $|x| \leq 1.0$, and 0 otherwise; and (f) $\gamma(x) = -x^2 + 1.0$ for $|x| \leq 1.0$, and 0 otherwise. D_F is set to be $\frac{1}{2}\lambda$ and λ for the left and right columns in Fig. 2, respectively. To draw contrast curves, 200 points are calculated in the range of spatial frequency from 0 to 2. The variation and sharp drops in the contrast curves at low spatial frequency such as in Figs. 2(a) and 2(d) are due to sharp boundaries of the pupil and characteristics of effective sources used.

We also calculated the image contrast for the harmonics higher than the second order. Features of the image contrast curves for odd and even harmonics of the higher order are similar to those for the fundamental and second harmonic spatial frequency components, respectively. However, values of the image contrast for higher harmonics become smaller as the order of harmonics increases. From these facts, the fundamental and second harmonic spatial frequency components play important roles to describe the outlines of image structures. To avoid excessive complexity, we only show the image contrast for these two components. The image contrast for the second harmonics also serves to

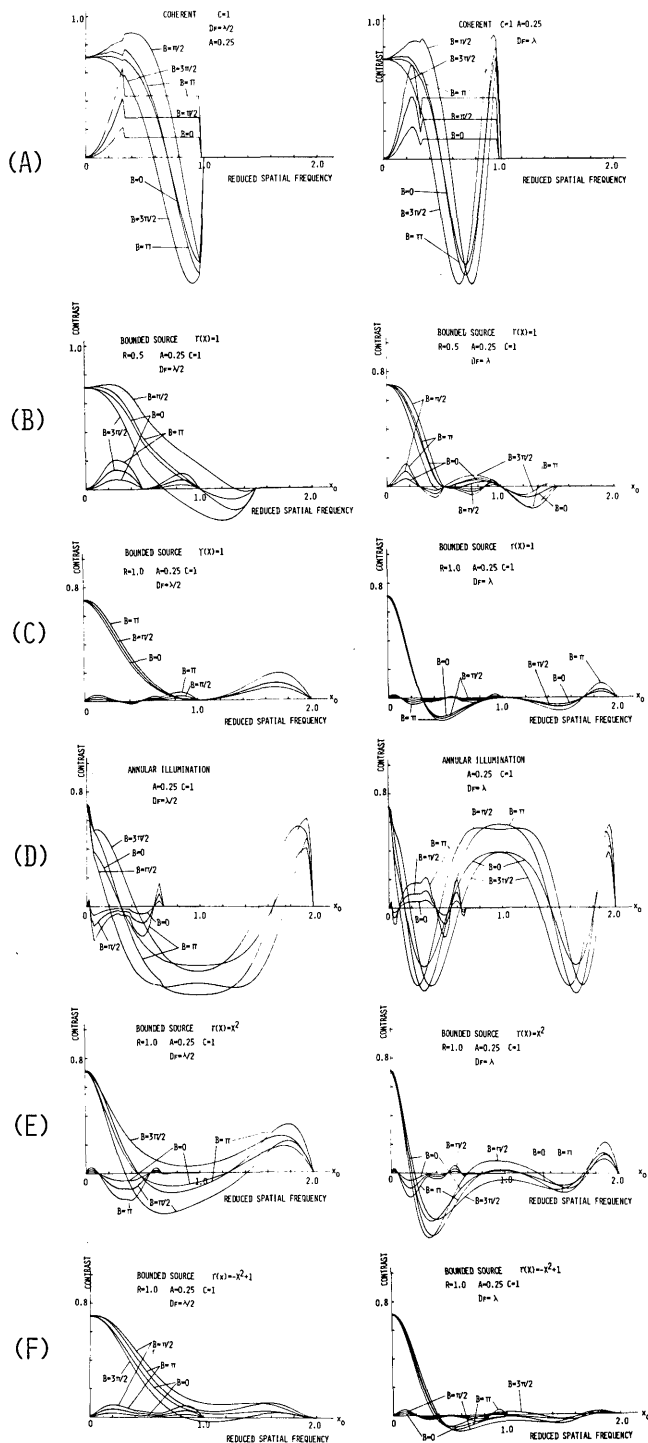


FIG. 2. Effects of phase variations in the square gratinglike complex objects on the image contrast for the fundamental and second harmonic spatial frequency components in defocused optical systems under six different illumination modes. Curves in the left column are obtained when the amount of defocusing is $D_F = \frac{1}{2}\lambda$, and those in the right column when $D_F = \lambda$. Light sources used to obtain images are (a) coherent source; (b) bounded source with uniform radiance, where source size is $R = 0.5$; (c) bounded source with uniform radiance, where source size is $R = 1.0$; (d) annular source [$\gamma(x) = 1.0$ for $0.95 \leq |x| \leq 1.0$, and 0 otherwise]; (e) bounded source with nonuniform radiance [$\gamma(x) = x^2$ for $|x| \leq 1.0$, and 0 otherwise]; (f) bounded source with nonuniform radiance [$\gamma(x) = -x^2 + 1.0$ for $|x| \leq 1.0$, and 0 otherwise].

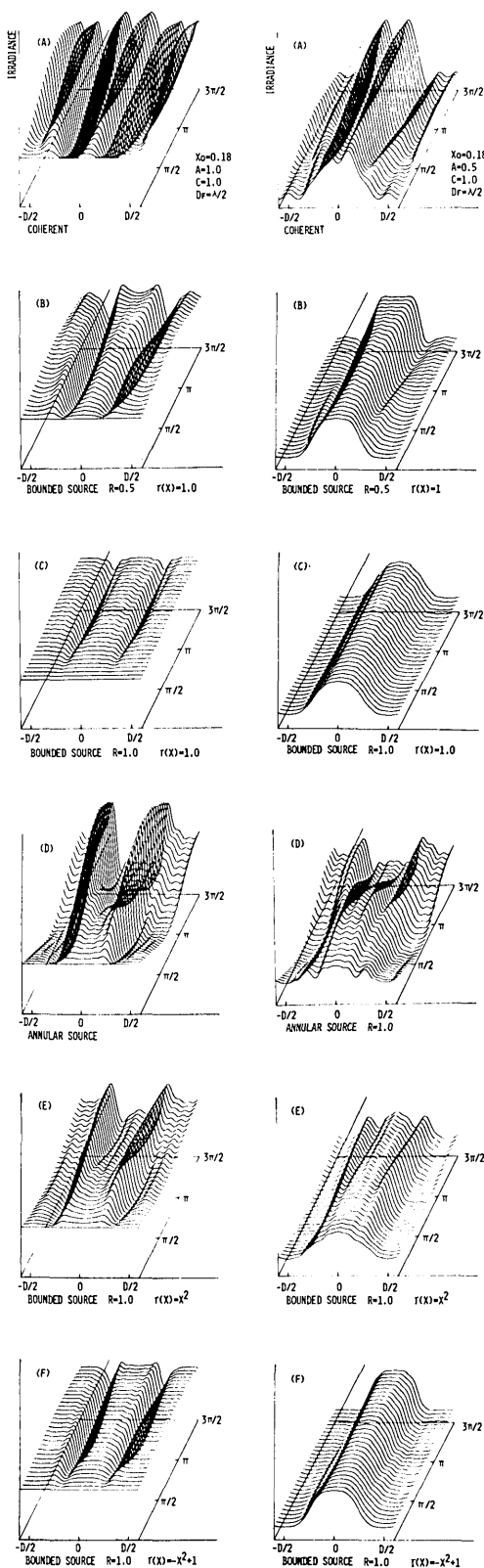


FIG. 3. Variations of appearances in defocused images of square gratinglike complex objects that have various phase variations. Light sources used to form images in (a)–(f) correspond to those in (a)–(f) in Fig. 2, respectively. The amount of defocusing is set to be $D_F = \frac{1}{2}\lambda$. Images in the left column are for pure phase objects [$A=C=1$ in Fig. 1(a)], and those in the right column are for semitransparent objects [$A=0.25$ and $C=1$ in Fig. 1(a)].

specify the nonlinearity due to a nonlinear system, because, for a square gratinglike complex object, the ideal image does not contain any harmonics of the even order. As shown in Fig. 2, the image contrast for the defocused image shows interesting behavior in accordance with a change in illumination condition and phase distribution in a complex object.

II. EVALUATION OF IMAGE APPEARANCE

A. One-dimensional image

Although the image contrast with respect to the fundamental and second harmonic spatial frequency components describes the outline of characteristics of the image formed by a defocused optical system under condition of partial coherence, it is also important to know effects of amplitude transmittance and phase variation on the appearance of images. To examine these effects, we simulate the images of square gratinglike complex objects and trapezoidal complex ones that have various object parameters. The fundamental spatial frequency x_0 of these objects are selected as 0.18. Then, the resultant image consists of harmonics up to thirteenth order.

Figures 3 and 4 show effects of phase variations on the images of the square gratinglike complex objects formed by a defocused optical system illuminated with effective sources specified by six configurations of radiance distribution used in Sec. I. The amount of defocusing is set to be $D_F = \frac{1}{2}\lambda$ for Fig. 3 and $D_F = \frac{1}{4}\lambda$ for Fig. 4. Figure 5 indicates images for trapezoidal complex objects when $D_F = \frac{1}{2}\lambda$, in which object parameters are set to be $s_1=0.1$, $s_2=0.31$, $s_3=0.5$, and $s_4=0.81$ in Fig. 1(b). For individual displays in Figs. 3–5, phase of the original object is changed along z axis from $B=0$ (pure amplitude object) to $\frac{3}{2}\pi$ by the interval of $\frac{1}{20}\pi$. In Figs. 3–5, images on the left- and right-hand sides are those for a pure phase object ($A=1.0$ and $C=1.0$) and semitransparent complex object ($A=0.5$ and $C=1.0$), respectively. The image appearance is also strongly affected by a change in the illumination mode, and fidelity of the image is getting worse in any illumination increase as the amount of defocusing and phase variation increase. As shown in Fig. 5, phase variation in a trapezoidal gratinglike complex object is extremely important to the image appearance in a defocused optical system under condition of partial coherence, which has not been encountered in partially coherent imaging due to an aberration-free optical system.²

Figure 6 shows enlarged images of the pure phase object (kinoform) under central coherent illumination at three different image planes, which are photographed by a camera lens ($f=50$ mm and $F=8$).

It seems to be strange that a pure phase object is highly visible in coherent image formation. The reason why an image of a pure phase object that has sharp boundaries of phase, formed by a diffraction-limited optical system under coherent illumination, has high contrast structures was discussed in detail in the previous paper.² The phenomenon could be interpreted by Gibbs phenomenon in the Fourier integral theory and by

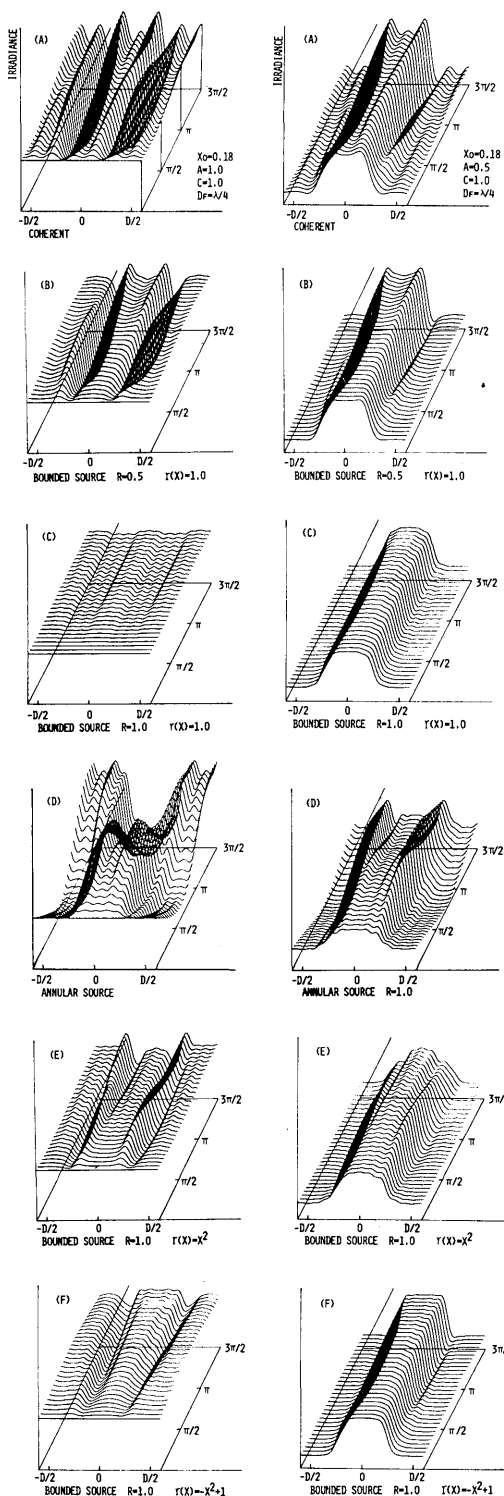


FIG. 4. Same as Fig. 3 but the amount of defocusing is $D_F = \frac{1}{4}\lambda$.

the effect of the negative amplitude caused by phase step in the object. To illustrate such phenomenon, we consider coherent imaging of the pure phase object whose complex transmittance is given by the function

$$O(u) = \begin{cases} 1, & u > 0, \\ 0, & u = 0, \\ e^{i\pi} (= -1), & u < 0. \end{cases} \quad (2)$$

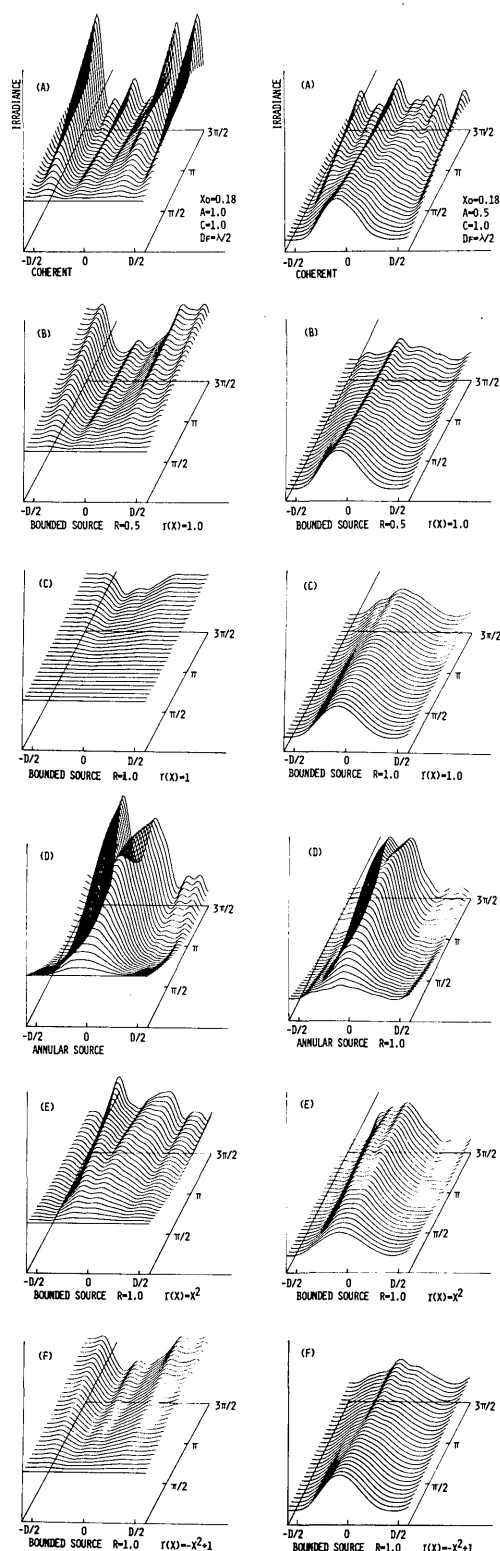


FIG. 5. Same as Fig. 3 but of trapezoidal gratinglike complex objects, where $D_F = \frac{1}{2}\lambda$.

Profile of this object is shown in broken lines in Fig. 7. Figures 7(a) and 7(b) show the complex amplitude and irradiance in the image of this edge object, respectively, which is formed by a diffraction-limited aberration-free optical system under central coherent illumination. A generation of irradiance dip in Fig. 7(b) at the loca-

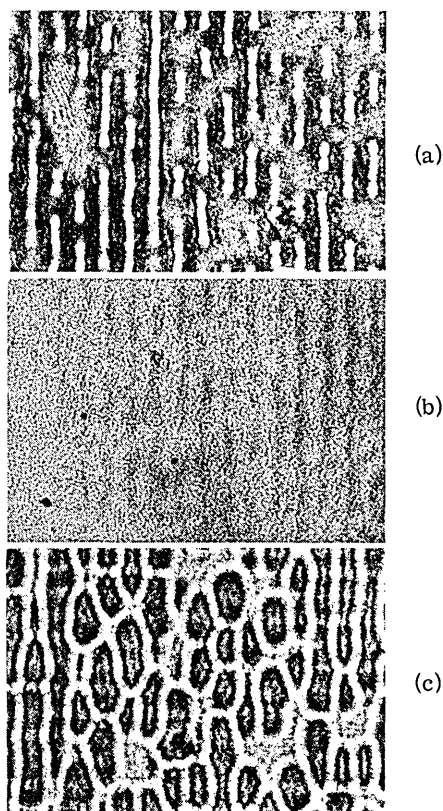


FIG. 6. Images of a pure phase object focused in three different image planes under coherent illumination. (a) Image in the best focal planes, (b) and (c) defocused images in different image planes.

tion of the sharp boundary indicates that such sharp phase step in a pure phase object becomes highly visible under coherent illumination. This phenomenon necessarily occurs in coherent imaging due to a defocused optical system.

Contrast reversal and variation of structures in the images in Figs. 6(a)–6(c) may be qualitatively interpreted by comparing values of the image contrast for the fundamental spatial frequency components. For example, the image contrast at spatial frequency $x_0 = 0.6$ in the case of $B = \pi$ in Fig. 2(a) has positive value when $D_F = \frac{1}{2}\lambda$, but it goes down as the amount of defocusing increases and shows negative value when $D_F = \lambda$. Such change of the image contrast seems to correspond to

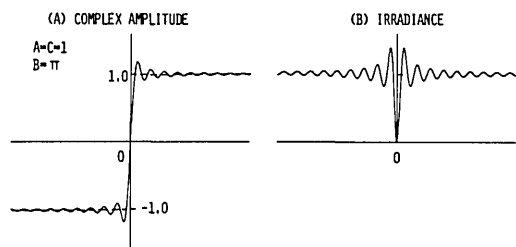


FIG. 7. Image of the edge object (pure phase object) that has phase step of π . (a) Complex amplitude in the image, and (b) image irradiance.

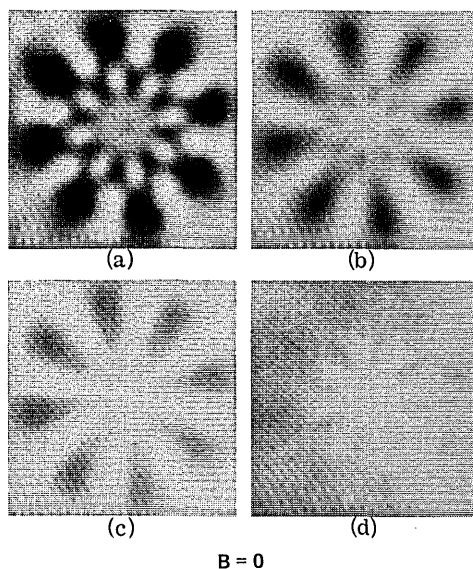


FIG. 8. Simulated images of the Siemens stars in the defocused optical system ($D_F = \frac{1}{2}\lambda$). Illumination conditions are (a) coherent illumination, (b) partially coherent illumination with the bounded source with uniform radiance ($R = 0.5$), (c) partially coherent illumination with the bounded source with uniform radiance ($R = 1.0$), (d) annular illumination. Object complex transmittance is $\frac{1}{2}$ for dark parts and 1.0 for bright ones (pure amplitude object).

variations of structures and contrast in the images in Figs. 6(a)–6(c).

From above facts, it should be noted that, in coherent imaging, the structure and contrast in the image of the complex object strongly depend on variations of the amount of defocusing, and the sharp boundaries of phase become highly visible. A generation of such irradiance dips is clearly encountered in two simulated images on the top of Figs. 3 and 4.

B. Two-dimensional image

Although one-dimensional images simulated serve to evaluate effects of phase variations on a change in image structures in a partially coherent optical system, it is still insufficient to assess the characteristics of the two-dimensional image of the complex object consisting of various spatial frequency components. To evaluate nonlinearity due to a nonlinear system in the presence of defocusing and nonlinear modulation of an object, it is better to simulate images of a two-dimensional complex object. According to the traditional method, we choose the Siemens star as the object model, which has amplitude transmittance and phase variation.

Figures 8–10 show simulated images of Siemens stars with various complex transmittances formed by a defocused optical system with four different illumination conditions. Numerical calculation has been made by use of the algorithm of the two-dimensional fast Fourier transform. The simulated images are displayed on a cathode ray tube in the digital image processor¹⁰ by 128×128 sampling points. Object illumina-

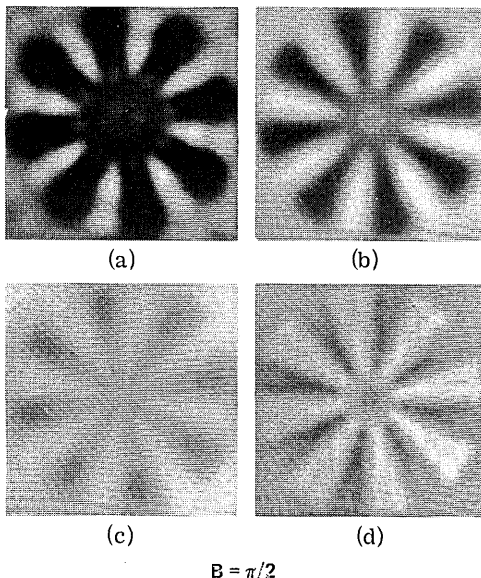


FIG. 9. Same as Fig. 8 but object complex transmittance is $\frac{1}{2} \exp(\frac{1}{2}i\pi)$ for dark parts and 1.0 for bright ones (complex object).

tion to obtain images in Figs. 8–10 are (a) coherent illumination, (b) partially coherent illumination with light radiated from the circular source with uniform radiance ($R=0.5$), (c) same as (b) but $R=1.0$, and (d) annular illumination, respectively, where R means the ratio of the condenser numerical aperture to the objective numerical aperture, which specifies coherence condition. The complex transmittance of the Siemens star is $e^{iB}/2$ for dark portions and 1.0 for bright ones. Phase variations are determined to be $B=0$ (pure amplitude object), $\frac{1}{2}\pi$, and π for Figs. 8, 9, and 10, respectively. The spatial frequency at the rim of the Siemens star is 0.3, and the amount of defocusing is set to be $D_F = \frac{1}{2}\lambda$ through these simulations.

It becomes evident from Figs. 8–10 that (i) the image appearance depends considerably upon the illumination mode and phase variations in an object, (ii) contrast reversal in simulated images occurs around contrast reversal frequencies in the image contrast curve for the fundamental spatial frequency component, and (iii) characteristics of the simulated two-dimensional image can be roughly estimated from behavior of the image contrast for the fundamental frequency component, because the image contrast for higher harmonic components is much less than that for the fundamental spatial frequency component. (In case of $x_0 > 0.3$, the resultant image consists of harmonics of the order less than the seventh.)

In Figs. 8–10, we also encounter the typical phenomena due to coherent and near coherent illumination, that is, dark parts in the image become broader than bright portions, and contrast reversed images with high contrast appear up to the high spatial frequency region. We also encounter peculiar appearance in the image under annular illumination, from which we can no more withdraw the real structures presented in the original object.

III. SUMMARY

By use of a technique to obtain the image of a periodic complex object in a partially coherent optical system and use of the fast Fourier transform algorithm, we calculated images and image contrast for one-dimensional periodic complex object, and then we simulated images of the Siemens star that had complex transmittance. The results presented here and further numerical results clarify a number of implications of combined effects of amplitude transmittance and phase variation in a complex object on appearances in the images obtained by a defocused optical system under partially coherent illumination.

(a) In a defocused optical system under coherent and near coherent illumination, the image contrast for the fundamental spatial frequency component maintains high level in the low-spatial-frequency region, and decays rapidly near the first contrast reversal frequency. On the other hand, under the condition of partial coherence, it decreases monotonically from zero spatial frequency to the first contrast reversal frequency.

(b) In coherent and near coherent illumination, the decreasing rate of the first contrast reversal frequency for the fundamental spatial frequency component with increasing the amount of defocusing is less than that in partially coherent illumination specified by an extended effective source. In other words, variation of image contrast with increasing the amount of defocusing is less sensitive in coherent, and near coherent illumination.

(c) In partially coherent imaging of the square gratinglike complex object, the image contrast with respect to even orders is regarded as the measure to specify the nonlinearity due to a nonlinear system. Then, the image contrast for the second harmonic component serves to describe image deformation to some extent.

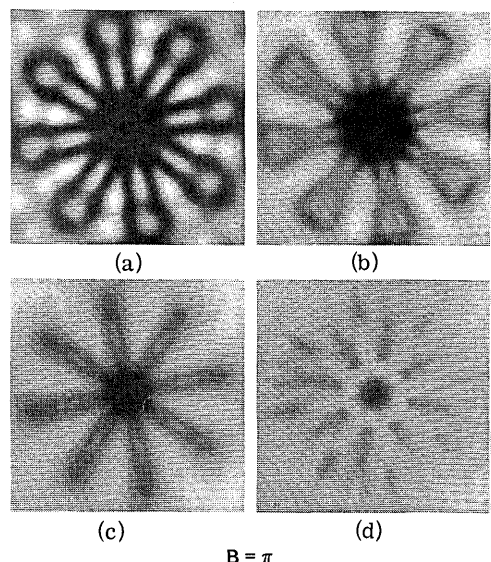


FIG. 10. Same as Fig. 8 but object complex transmittance is $\frac{1}{2} \exp(i\pi)$ for dark parts and 1.0 for bright ones (complex object).

The image contrast for the second harmonics in coherent and near coherent illumination is higher than that in a partially coherent optical system illuminated with light radiated from extended effective sources. Thus, if an original object consists of a wide range of spatial frequency components, fidelity of the image becomes worse in coherent and near coherent illumination.

(d) Although phase variation in an object is extremely influential for a change in appearance in the image formed by an aberration-free optical system, it is less influential in the presence of large defocusing.

(e) A defocused optical system is very influential for variation of structures in the image of the complex object, that has continuous phase variations. This phenomenon is one of typical effects of defocusing on partially coherent imaging of a complex object, because it was not encountered in imaging due to an aberration-free optical system.²

(f) Variation of radiance distribution in the effective source affects behavior of the contrast curve. This suggests that manipulation of the radiance in an illuminating light source offers useful technique in improving fidelity of the image of a complex object, formed by a defocused optical system in partially coherent illumination.

(g) Insofar as we consider partially coherent imaging of the complex object that has proper amplitude contrast, the image contrast of the fundamental spatial frequency component can describe outline of the characteristics of the resultant image well. Of course, this is not valid for the low contrast object that has large phase variations.

Results in this paper and two previous papers, and further numerical results, reveal significant information about partially coherent imaging of a complex object. When the original object has amplitude transmittance and phase variation, the object spectrum consists of a number of spatial frequency components. This makes it more difficult to treat partially coherent imaging of a complex object than to do so for the pure amplitude

object. However, when we consider the image formation in a practical optical system, the object or specimen contains necessarily phase variation as well as variation of amplitude contrast. Thus, the problem to clarify the combined effects of them on partially coherent imaging is inevitable for the experimenters who would observe the small specimen in a microscope and measure photographic density in emulsion with microdensitometer. To clarify all phenomena concerned with the stated effects is impractical for us, because of nonlinearity due to nonlinear system and nonlinear modulation of an object. To do so, phenomena under conditions specified by combination of a number of parameters describing object and system characteristics must be assessed. Of course, results in this paper are insufficient to the stated problem fully but they have been useful to evaluate characteristics and outlines of partially coherent imaging of a complex object. Hence, results in this paper are set up to answer the stated problem and point out the direction that we should examine in the next step. The computer algorithm and analytical procedure are also applicable to investigate further study on this problem.

In conclusion, although the image assessment for the complex object in partially coherent optical system is extremely difficult, general properties which we obtained have to be taken fully into account in microscopy and in microdensitometry.

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